

1. Information Exchange analysis of a cache-blocked matrix-vector multiply

Consider the dense matrix-vector multiplication

$$\vec{y} \leftarrow A \cdot \vec{x},$$

where A is a matrix of size N^2 . Use double-precision numbers.

Implement a cache-blocked version of this numerical task with block size B with $N \pmod B = 0$. Compute the information exchanged with external memory I_{mem} as a function of N and B under the following assumptions:

- The entire vector $\vec{x}$ cannot be kept in cache.
- A fraction N/B of the vector can stay in cache.

2. Cachegrind

Use Cachegrind to model the number of cache misses for the cache-blocked matrix-vector multiply routine as a function of B . Choose N sufficiently large, such that the vector does not fit in the L1D cache.

3. Memory allocation overhead

Design a benchmark for measuring the overhead of allocating and freeing memory dynamically. Determine the time needed for a single memory allocation and free pair. Justify your approach to obtain sufficiently precise results. Optionally also test the dependence on the amount of allocated memory.